

BANK CUSTOMER CHURN PREDICTION

End-to-End Data Science Project

EDA · SQL Data Warehouse · Machine Learning · Dashboard

10,000

Customers

5 Phases

Completed

XGBoost

Best Model

0.869

AUC Score

Tech Stack: Python · pandas / numpy · scikit-learn · XGBoost / LightGBM · SHAP · MySQL · HTML/CSS/JS

Executive Summary

This project delivers a complete churn prediction system for a European retail bank with 10,000 customers across France, Germany, and Spain. The objective was to identify which customers are likely to close their accounts, understand the behavioural drivers behind that decision, and produce a model and dashboard that a business team can act on immediately.

The dataset presented a classic 80/20 class imbalance (20.4% churn rate), zero missing values, and one potential leakage feature — the *Complain* column — which was quarantined throughout the pipeline and excluded from all model training.

Project KPIs

20.4%	32.4%	55.3%	100%	0.869
Overall Churn Rate	Germany Churn Rate	Peak Age Band (48-58)	Products 4 Churn Rate	XGBoost AUC

Key Findings

Germany	32.4% churn rate — nearly twice that of France (16.2%) and Spain (16.7%). German female customers are the single highest-risk segment at 37.6%.
Age	Non-linear churn curve peaking sharply in the 48–58 band (55.3%). Youngest customers (18–28) show only 7.2% churn.
Products	Customers with 3 or 4 products churn at 82.7% and 100% respectively — the strongest single predictor in the dataset. Products 1 and 2 behave normally.
Balance	Counter-intuitive: customers with a positive balance churn more (24.1%) than those with zero balance (13.8%). Higher balance may reflect disengagement.
Activity	Active members churn at 14.3% vs 26.9% for inactive members. Engagement programmes have a measurable protective effect.
Complain	Excluded from all modelling as a leakage feature. Available in the cleaned layer for post-hoc business analysis only.

Methodology

Phase 1 — EDA

Exploratory analysis across 18 features. Distribution plots, categorical churn rates, correlation heatmap, interaction effects (Geography x Gender, Products x Activity), balance analysis, and age band segmentation. Leakage investigation on the Complain column.

Phase 2 — SQL Data Warehouse

Three-layer MySQL schema: staging_raw (raw ingest) → customers_cleaned (validated, typed, duplicates removed) → customer_features (30 engineered features). Includes 7 analytical views for BI consumption. Complain quarantined at the cleaned layer.

Phase 3 — Machine Learning

Logistic Regression baseline, XGBoost, and LightGBM trained on an 80/20 stratified split. Class imbalance handled via class_weight=balanced and scale_pos_weight. Evaluated on ROC-AUC, F1, and 5-fold cross-validation. SHAP values computed for explainability.

Phase 4 — Dashboard

Interactive HTML/CSS/JS dashboard with animated KPI cards, bar charts, donut chart, age-band visualisation, Geography x Gender heatmap, and model comparison panel. Deployable to GitHub Pages.

Phase 5 — Report & GitHub

Model card, executive summary PDF, structured README, and full GitHub repository with reproducible folder layout.

Model Results

Three models were evaluated. XGBoost achieved the best performance on both ROC-AUC and F1. The cross-validation AUC closely tracks the test AUC, indicating the model generalises well with no significant overfitting.

Model	ROC-AUC	F1 Score	CV-AUC (5-fold)
XGBoost ★	0.8688	0.6230	0.8525
LightGBM	0.8669	0.6168	0.8513
Logistic Regression	0.8500	0.5776	0.8342

All models used class imbalance correction. Logistic Regression used class_weight="balanced"; XGBoost used scale_pos_weight=3.91 (ratio of negative to positive class). Features were standardised for Logistic Regression via StandardScaler inside a Pipeline.

Feature Engineering

All features mirror the SQL feature mart (Phase 2), ensuring consistency between the data warehouse and the ML pipeline. 30 features were used in total.

Feature Group	Engineered Features	Rationale
Age	age_band, is_peak_churn_age (48-58)	Non-linear churn curve; 48-58 band = 55.3%
Balance	has_zero_balance, log_balance, balance_to_salary_ratio	Zero balance = lower churn; log transform for skew
Products	is_high_risk_products (>=3), is_single_product	Products 3&4 near-100% churn; strongest predictor
Tenure	is_new_customer (tenure=0), products_per_year	New customers show different risk profile
Engagement	engagement_score (0-3), is_disengaged	Composite of activity + card + satisfaction
Geography	geography_risk_score, is_high_churn_geography	Germany=3, Spain=2, France=1 ordinal encoding
Interactions	is_german_female, is_inactive_high_products	Captures joint effects from EDA heatmaps

Model Card

Model type	XGBClassifier (XGBoost 3.2.0)
Task	Binary classification — customer churn prediction
Training data	8,000 rows (80% stratified split), 30 features
Test data	2,000 rows (20% stratified split)
Class handling	scale_pos_weight = 3.91 (negative/positive ratio)
Hyperparameters	n_estimators=400, max_depth=5, learning_rate=0.05, subsample=0.8, colsample_bytree=0.8
Metrics (test)	AUC=0.8688, F1=0.6230, CV-AUC=0.8525
Explainability	SHAP TreeExplainer — global importance + value distribution
Leakage controls	Complain column excluded from all training and inference
Limitations	Synthetic/academic dataset. Not validated on live bank data. Re-train before any production use.

Business Recommendations

- **Germany retention programme:** At 32.4% churn — nearly double other markets — Germany warrants a dedicated intervention. German female customers (37.6%) should be the first target cohort.
- **Product rationalisation review:** The 82–100% churn rate for customers with 3+ products is anomalous. Investigate whether these are bundled products the customer did not choose, or a data quality issue. Either way, this needs immediate attention.
- **Age-based engagement:** The 48–58 cohort (55.3% churn) is likely approaching retirement and reassessing their financial relationships. Targeted wealth management or advisory outreach could meaningfully reduce this rate.
- **Reactivation campaign:** Inactive members churn at 26.9% vs 14.3% for active members — an 88% relative difference. Even modest reactivation (login incentives, product check-ins) has measurable ROI.
- **High-balance early warning:** Customers with positive balances churn more than zero-balance customers. This suggests the model should flag high-balance inactive customers for priority outreach before they withdraw and leave.